OSN2526: Assignment 2: Page tables

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1 Inleiding

This report describes an analysis of different virtual memory implementations. In Section 2 a comparative analysis is performerd between a simple single level MMU and a RISC-V Sv48 MMU. Both MMU are simulated within the provided framework given for the assignment.

2 MMU Comparison

This section performes a comparative analysis of a simple single level MMU provided for the assignment and a RISC-V Sv48 MMU implementation.

Table 1 shows the results for a single process.

Tabel 1: Single process results of RISC-V Sv48 and Simple MMU Implementations for trace: gcc.txt.gz.

Metric	**RISC-V (Sv48)**	**Simple (Single-Level)**
Page size	4 KB	64 MB
Page table allocated	36 KiB	64 MiB
Handled page faults	6	4
Max. # allocated physical pages	15	5

Tabel 2: Multiprocess results of RISC-V Sv48 vs. Simple MMU (gcc.txt.gz and ls.txt.gz)

Metric	**RISC-V (Sv48)**	**Simple (Single-Level)**
Page size	4 KB	64 MB
Page table allocated	68 KiB	128 MiB
Handled page faults	11	7
Max. # allocated physical pages	28	8

3 Conclusie